Dynamic Model of the Ball-Bat Collision

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This document describes the computational technique used by the program *ACOR* to calculate the collison between ball and bat. It is an update of the work published in American Journal of Physics **68**, 979-990 (2000).

I. DYNAMIC MODEL FOR THE BAT

In Fig. 1, a baseball bat that has been distorted from its equilibrium shape due to bending and shearing is schematically shown. The distortion of a particular segment of the bat can be characterized by its transverse displacement $y(z)$ and orientation $\phi(z)$ as a function of its coordinate z along the long axis of the bat. The bat is divided into N parallel slices of circular cross section, each of thickness Δz and each elastically coupled to its neighbors through the Young's modulus Y and the shear modulus S. The equations of motion of y and ϕ for free (i.e., no external force or torque) vibrations of a bat with nonuniform cross section were derived by VanZandt[1], rederived and slightly corrected by Nathan [2], and again rederived in a more symmetric form herein:

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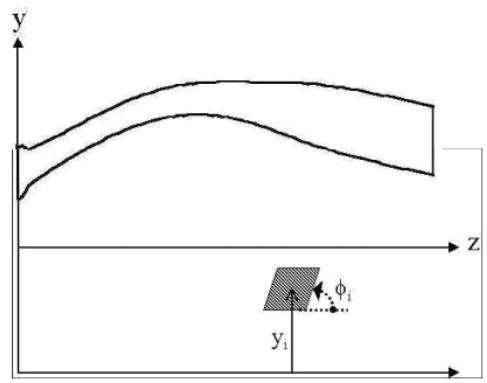

FIG. 1: Schematic (greatly exaggerated) representation of a bat that is bent and sheared. The coordinate y_i is the displacement of the ith section of the bat relative to the z axis, whereas ϕ_i is the angle between the plane of the ith section and the y axis.

$$\ddot{y}_i = \frac{S_i}{\rho_i \Delta z^2}\left[(y_{i+1} + y_{i-1} - 2y_i) + \left(\frac{S_i A_i - S_{i-1}A_{i-1}}{S_i A_i}\right)\left(y_i - y_{i-1} + \frac{\Phi_i + \Phi_{i-1}}{2}\right) + \frac{\Phi_{i+1} - \Phi_{i-1}}{2}\right] \tag{1}$$

and

$$\begin{aligned}\ddot{\Phi}_i &= \frac{Y_i}{\rho_i \Delta z^2}\left[(\Phi_{i+1} + \Phi_{i-1} - 2\Phi_i) + \frac{Y_i I_i - Y_{i-1}I_{i-1}}{Y_i I_i}(\Phi_i - \Phi_{i-1})\right] \\ &- \frac{1}{2\rho_i I_i}\left[S_i A_i\left(y_{i+1} - y_{i-1} + \frac{2\Phi_i + \Phi_{i+1} + \Phi_{i-1}}{2}\right)\right. \\ &- \left.(S_i A_i - S_{i-1}A_{i-1})\left(y_i - y_{i-1} + \frac{\Phi_i + \Phi_{i-1}}{2}\right)\right],\end{aligned} \tag{2}$$

where ρ, A_i, and I_i are the density, cross sectional area, and area moment of inertia of slice i (about an axis passing through the center of mass of the

slice and normal to the y-z plane), respectively, and $\Phi_i = \phi_i \Delta z$. The above equations are supplemented with boundary conditions, which we take to be those of a bat that is free on both ends. Therefore the force and torque on the end slices are due only to the adjacent inner slices. Note that these equations simplify if the elastic constants, Y_i and S_i are independent of i.

Note that these equations are the discrete form of the so-called Timoshenko beam equations[3], suitably modified for a beam of non-uniform ρ, A, I, Y, and S. All the bats considered herein have a circular cross section. For solid bats (such as a typical wood bat), a slice of radius R_i has $A_i = \pi R_i^2$ and $I_i = \pi R_i^4/4 = A_i R_1^2/4$. For hollow bats (such as a typical aluminum bat), a slice of inner radius $R_{1,i}$ and outer radius $R_{2,i}$ has $A_i = \pi(R_{2,i}^2 - R_{i,1}^2)$ and $I_i = \pi(R_{2,i}^4 - R_{1,i}^4)/4$.

In order to find the normal modes of the bat implied by the equations of motion, Eqs. 1 and 2 are first rewritten in compact matrix notation:

$$\ddot{\psi} = \mathbf{H}\psi \,, \tag{3}$$

where ψ is a 2N-element column matrix

$$\psi \equiv \begin{pmatrix} y_1 \\ \vdots \\ y_N \\ \Phi_1 \\ \vdots \\ \Phi_N \end{pmatrix} \equiv \begin{pmatrix} \vec{y} \\ \vec{\Phi} \end{pmatrix} \tag{4}$$

and $\mathbf{H}$ is a *nonsymmetric* 2N x 2N matrix.

From Eqs. 1-2 one can find the elements of **H**. To simplify the notation, we first define

$$\Lambda \equiv \frac{S}{\rho \Delta z^2} \quad \Upsilon \equiv \frac{Y}{\rho \Delta z^2}$$

Then for i=2...N-1 and j=i+N, we have the following:

$$H_(i, i+1) = \Lambda \qquad H_{i,i} = -\Lambda(1 + \frac{A_{i-1}}{A_i})$$

$$H_(i, i-1) = \Lambda \frac{A_{i-1}}{A_i} \qquad H_{i,j+1} = \frac{\Lambda}{2}$$

$$H_{i,j} = \frac{\Lambda}{2}(1 - \frac{A_{i-1}}{A_i}) \qquad H_{i,j-1} = -\Lambda \frac{A_{i-1}}{2A_i}$$

$$H_{j,j+1} = \Upsilon - \Lambda \frac{A_i \Delta z^2}{4I_i} \qquad H_{j,j} = -\Upsilon(1 + \frac{I_{i-1}}{I_i}) - \Lambda \frac{A_i \Delta z^2}{4I_i}(1 + \frac{A_{i-1}}{A_i})$$

$$H_{j,j-1} = \Upsilon \frac{I_{i-1}}{I_i} - \Lambda \frac{A_{i-1} \Delta z^2}{4I_i} \qquad H_{j,i+1} = -\Lambda \frac{A_i \Delta z^2}{2I_i}$$

$$H_{j,i} = \Lambda \frac{A_i \Delta z^2}{2I_i}(1 - \frac{A_{i-1}}{A_i}) \qquad H_{j,i-1} = \Lambda \frac{A_{i-1} \Delta z^2}{2I_i}$$

For the end pieces i=1,N and j=i+N, we assume a bat that is free at both ends, so that the forces and torques on the end slices are due only to the one

adjacent slice. We have the following:

$$H_{1,2} = \Lambda \qquad H_(1,1) = -\Lambda$$

$$H_{1,N+2} = \frac{\Lambda}{2} \qquad H_{1,N+1)} = \frac{\Lambda}{2}$$

$$H_{N,N} = -\Lambda\frac{A_{N-1}}{A_N} \qquad H_{N,N-1} = \Lambda\frac{A_{N-1}}{A_N}$$

$$H_{N,2N-1} = -\Lambda\frac{A_{N-1}}{2A_N} \qquad H_{N,2N} = -\Lambda\frac{A_{N-1}}{2A_N}$$

$$H_{N+1,N+1} = \Upsilon - \Lambda\frac{A_N\Delta z^2}{4I_N} \qquad H_{N+1,N+2} = -\Upsilon - \Lambda\frac{A_N\Delta z^2}{4I_N}$$

$$H_{N+1,1} = \Lambda\frac{A_N\Delta z^2}{4I_N} \qquad H_{N+1,2} = -\Lambda\frac{A_N\Delta z^2}{4I_N}$$

$$H_{2N,2N-1} = \Upsilon\frac{I_{N-1}}{I_N} - \Lambda\frac{A_{N-1}\Delta z^2}{4I_N} \qquad H_{2N,2N} = -\Upsilon\frac{I_{N-1}}{I_N} - \Lambda\frac{A_{N-1}\Delta z^2}{4I_N}$$

$$H_{2N,N-1} = \Lambda\frac{A_{N-1}\Delta z^2}{2I_N} \qquad H_{2N,N} = -\Lambda\frac{A_{N-1}\Delta z^2}{2I_N}$$

All other matrix elements are identically zero.

Harmonic vibrations are assumed,

$$\ddot{\psi} = -\omega^2\psi\,. \tag{5}$$

With some manipulation of Eqs. 1 and 2, $\mathbf{H}$ can be transformed into the form $\mathbf{H} = \mathbf{M}^{-1}\mathbf{K}$, where $\mathbf{K}$ is a *symmetric* 2N x 2N "stiffness" matrix[4], and $\mathbf{M}$ is a *diagonal* mass matrix. The elements M_{kk} of the mass matrix are the mass of the kth slice $\rho A_k\Delta z$ for $1 \leq k \leq N$ and $\rho I_k/\Delta z$ for $N < k \leq 2N$. Clearly then, $\mathbf{K} = \mathbf{MH}$. With these definitions, the equations can be written in the form

$$\mathbf{K}\psi_n \;=\; -\omega_n^2\mathbf{M}\psi_n\,, \tag{6}$$

which is immediately recognized as a generalized eigenvalue equation, requiring the diagonalization of $\mathbf{K}$ in order to find the normal mode frequencies ω_n and

associated eigenvectors ψ_n.

Eq. 6 can be further simplified using the so-called Cholesky Decomposition.[5] We define the symmetric matrix $\mathbf{B} = \mathbf{M^{-1/2}KM^{1/2}}$, so that $B_{ij} = K_{ij}/\sqrt{M_i M_j}$. Then after a bit of manipulation we arrive at

$$\mathbf{B}(\mathbf{M^{1/2}}\psi_n) \;=\; -\omega_n^2(\mathbf{M^{1/2}}\psi_n)\,. \tag{7}$$

Eq. 7 is now a simple eigenvalue equation, where the eigenvalues are identical to those of the original matrix $\mathbf{H}$ and the eigenvectors $B_{ij} = H_{ij}/\sqrt{M_i}$.

Standard numerical techniques are used to accomplish this.[6] The lowest two modes are zero-frequency rigid body modes corresponding to uniform translation (y_{ni} independent of i, Φ_{ni}=0) and uniform rotation (y_{ni} linear in i, Φ_{ni} independent of i). The eigenvectors are normalized and mutually orthogonal

$$\sum_i \left(A_i y_{mi} y_{ni} + \frac{I_i}{\Delta z^2}\Phi_{mi}\Phi_{ni} \right) \Delta z \;=\; \bar{A}\delta_{mn} \tag{8}$$

which defines the parameter $\bar{A}$ and where the sum is over all the slices i.

II. THE BALL-BAT COLLISION

Thus far, we have only considered a free bat. For the ball-bat collision, an additional time-dependent force appears on the right-hand-side of Eq. 3

$$\ddot{\psi} \;=\; \mathbf{H}\psi \;+\; \mathbf{M}^{-1} \begin{pmatrix} \vec{F}(t) \\ 0 \end{pmatrix}\,, \tag{9}$$

where the ith element of the column vector $\vec{F(t)}$ is the time-dependent external force that the ball exerts on the ith slice of the bat. One way to solve the

collision problem is to solve Eq. 9 directly, using Runge-Kutta techniques to solve the coupled equations of motion for each slice. Alternately, since the normal modes of the free vibration form a complete set, the solution of Eq. 9 can be written as an expansion

$$\psi(t) \;=\; \sum_n a_n(t)\psi_n \,, \tag{10}$$

so that

$$\begin{aligned} \sum_n \left(\ddot{a}_n(t) + \omega_n^2 a_n(t)\right) \vec{y}_n \;&=\; \mathbf{M}^{-1}\vec{F}(t) \\ \sum_n \left(\ddot{a}_n(t) + \omega_n^2 a_n(t)\right) \vec{\Phi}_n \;&=\; 0 \end{aligned} \tag{11}$$

Using the orthogonality of the eigenstates, Eq. 8, to project out the mth mode, one finds

$$\ddot{a}_m(t) + \omega_m^2 a_m(t) \;=\; \frac{\sum_i F_i(t) y_{mi}}{\rho \bar{A}} \,, \tag{12}$$

where the sum on the right-hand-side is over all slices of the bat. Eq. 12 is a series of N simple harmonic oscillators, all coupled via the force F. If there is natural damping in the bat, then an additional term $\Gamma \dot{a}_m(t)$ appears on the LHS of Eq. 12, leading to exponential damping of the a_m in the form $e^{-\Gamma t/2}$. For so-called "stiffness-proportional Rayleigh damping"[7], $\Gamma = 2\omega_m/Q$, the exponential damping takes the form $e^{-\omega_m t/Q}$, where Q is a dimensionless constant and the decay time constant is therefore Q/ω_m. Van Zandt[1] suggests using Q=10, leading to a time constant of 5 ms for ω_m= 2000 sec^{-1} (f_m=318 Hz).

In the ball-bat collision, the driving force $F_i(t)$ is the force that the ball and bat mutually exert on each other. This force depends on the compression of

the radius of the ball, which is defined as the separation between the surface of the bat $y_k(t) = \sum_n a_n(t) y_{nk}$ and the center of the ball $y_{\rm ball}(t)$ but offset by the natural radius $R_{\rm ball}$. Therefore $u = 0$ corresponds to no compression:

$$u(t) = R_{\rm ball} - (y_{\rm ball}(t) - y_k(t)) \ , \tag{13}$$

where k is the slice of the bat where the impact occurs. The complete dynamics of the collision are contained in Eqs. 10, 12 and 13, the equation of motion of the ball

$$m_{\rm ball}\, \ddot{y}_{\rm ball} = -F(u(t)) \, , \tag{14}$$

the normal modes of the bat, the dependence of the force on u, and the initial conditions. For the latter, $t = 0$ is taken to be the time when the ball and bat come into contact for the first time ($u = 0$, $\dot{u} < 0$), at which time the ball and the rigid body modes of the bat each have initial velocities which must be specified. Under typical conditions, the ball and bat (at the impact point) are initially moving in opposite directions. Since the bat is assumed not to be vibrating prior to the collision, $a_n(0) = \dot{a}_n(0) = 0$ for the true vibrational modes.

Standard 4th-order Runge-Kutta[6] (RK4) is used to integrate the coupled differential equations numerically, using a time grid adjusted to give stable results. For a typical collision lasting less than 1 ms, a mesh of 5 μs is more than adequate. When the ball and bat separate ($u = 0$, $\dot{u} > 0$), the force goes to zero, the bat vibrates freely, and the ball exits with constant velocity v_f.

It might be helpful to elaborate on the implementation of the RK4 technique in the present context. Solving the coupled 2nd order differential equations

Eqs. 12 and 14 requires specifying the time derivatives of 2N+2 quantities, assuming that N normal modes are included in the calculation: $a_n(t)$ and $\dot{a}_n(t)$ for $n = 1...N$ and y_{ball} and $\dot{y}_{ball}$. One way to implement this is to form a 2N+2 element column matrix **A** as follows:

$$\begin{aligned} A_n &= a_n \\ A_{N+n} &= \dot{a}_n \\ A_{2N+1} &= y_{ball} \\ A_{2N+2} &= \dot{y}_{ball}\,, \end{aligned} \tag{15}$$

where $n = 1...N$. The equations of motion imply that the time derivatives are given by

$$\begin{aligned} \dot{A}_n &= a_{N+n} \\ \dot{A}_{N+n} &= \ddot{a}_n = -\omega^2 a_n + F y_{nk} \\ \dot{A}_{2N+1} &= \dot{y}_{ball} \\ \dot{A}_{2N+2} &= \ddot{y}_{ball} = -F/m_{ball}\,, \end{aligned} \tag{16}$$

where I have assumed that the ball-bat force acts on a single slice with label k. That is, the sum over slices on the RHS of Eq.12 has only a single term, with $i = k$. Alternately, given the finite size of the ball, one could spread the force over multiple slices centered at slice k; for example

$$F y_{nk} \rightarrow F\left(\frac{y_{n,k-2} + 4y_{n,k-1} + 9y_{n,k} + 4y_{n,k+1} + y_{n,k+2}}{19}\right) \tag{17}$$

Finally, it is useful to note that the normalization condition, Eq. 8, implies that the y_{mi} and ψ_{mi} have the dimensions [length$^{-0.5}$]. Since the LHS of Eq. 10 is necessarily a length, the a_m have dimensions [length$^{1.5}$].

A. Implementation of External Input

The formalism described above may be used to solve the equations of motion when the normal mode frequencies ω_n, eigenstates ψ_n, and mass matrix $\mathbf{M}$ are already known, perhaps from an experimental modal analysis or a finite-element analysis, assuming the ψ_n are normalized as per Eq.8. From a practical point of view, it is straightforward to determine the "upper half" of the ϕ_n from a modal analysis (i.e., the $\vec{y}_n$) but not as easy to determine the "lower half" (i.e., the $\vec{\Phi}_n$). Eq. 9 shows that the $\vec{\Phi}_n$ do not enter directly into the collision equations of motion, but they do enter indirectly into the normalization, Eq. 8, since the sum over upper and lower components in implied in the matrix multiplication. One possible procedure is to assume no shearing of the slices, so that $\Phi_n = -\frac{dy_n}{dz}\Delta z$. The accuracy of this technique needs to be checked numerically.

B. Frozen Degrees of Freedom

The calculations in Ref. [2] show that for a typical ball-bat collision, energy is transferred to only the lowest few modes. That is not to say the higher modes are not important to the collision, since they can contribute to a "trampoline-like" effect. One approximate technique is to treat these modes statically (hence, "frozen"), meaning that the time derivative in Eq. 12 is neglected so that

$$\omega_m^2 a_m(t) \approx F_k(t) y_{mk} \,.$$

Therefore the location of the surface of the bat at the impact location can be written

$$y_k(t) \;=\; \sum_{m=1}^{m_0} a_m(t) y_{mk} \;+\; F_k \sum_{m>m_0} \frac{y_{mk}^2}{\omega_m^2}\,,$$

where the modes in first sum (up to m_0) are calculated from explicit solution of Eq. 12 while the second sum is an approximate expression for the remaining frozen modes.

C. Pivoted Bat

Thus far, we have only considered bats that are totally free. However, when testing bats in a typical laboratory setting[8], a baseball is fired from a high-speed air cannon onto a stationary bat which is attached at the handle to a structure that allows it to pivot freely about a point. Such a bat is not free and the modifications to our formalism must be considered.

A sketch of the typical situation is shown in Fig. 2. The side view shows the pivoting structure at the handle. The top view shows the bat rotated by angle θ about the vertical pivot axis, which is perpendicular to the plane of the drawing. The bat is attached to the pivot by clamping it at two points symmetrically located a distance r_p on either side of the pivot axis. In a typical application, the pivot axis is six inches from the knob and r_p=1 inch. The effect of these clamps is to exert additional forces on the bat, f_1 and f_2, to constrain the slices of the bat at the clamps to rotate about the pivot axis. Denoting these two slices of the bat as 1 and 2 and their displacement as y_1

and y_2, the clamps constrain those slices so that

$$\ddot{y}_2 \;=\; -\ddot{y}_1 \;=\; r_p\ddot{\theta}\,. \tag{18}$$

On the other hand, considering the bat-plus-pivot as a single system, the only external force on that system is f_{ball}, the force exerted on the bat by the ball. That force exerts a torque on the system, resulting in the rotation about the pivot axis:

$$\ddot{\theta} \;=\; \frac{r_0 f_{ball}}{(I_0 + I_p)}\,, \tag{19}$$

where I_0 and I_p are the moments of inertia of the bat and pivot, respectively, about the pivot axis, and r_0 is the distance from the ball-bat impact point to the pivot axis. These two equations are sufficient to determine uniquely the constraining forces f_1 and f_2, as shown below.

From Eqs.10-12, we have

$$\begin{aligned}
\ddot{y}_1 \;&=\; \sum_n \left(-\omega_n^2 a_n y_{n1} \;+\; \frac{y_{n1}}{\rho\bar{A}}\left(f_{ball} y_{nk} \;+\; f_1 y_{n1} \;+\; f_2 y_{n2}\right)\right) \\
\ddot{y}_2 \;&=\; \sum_n \left(-\omega_n^2 a_n y_{n2} \;+\; \frac{y_{n2}}{\rho\bar{A}}\left(f_{ball} y_{nk} \;+\; f_1 y_{n1} \;+\; f_2 y_{n2}\right)\right)
\end{aligned} \tag{20}$$

where the subscript k denotes the slice where the impact with the ball occurs. From Eqs. 18-19, the LHS of the preceding two equations are

$$\begin{pmatrix} \ddot{y}_1 \\ \ddot{y}_2 \end{pmatrix} \;=\; \mp\,\frac{r_p r_0 f_{ball}}{(I_0 + I_p)}\,. \tag{21}$$

Eqs. 20 and 21 involve two equations and two unknowns, f_1 and f_2, which can easily be solved. These forces, along with f_{ball}, are the forces on the RHS of Eq. 12.

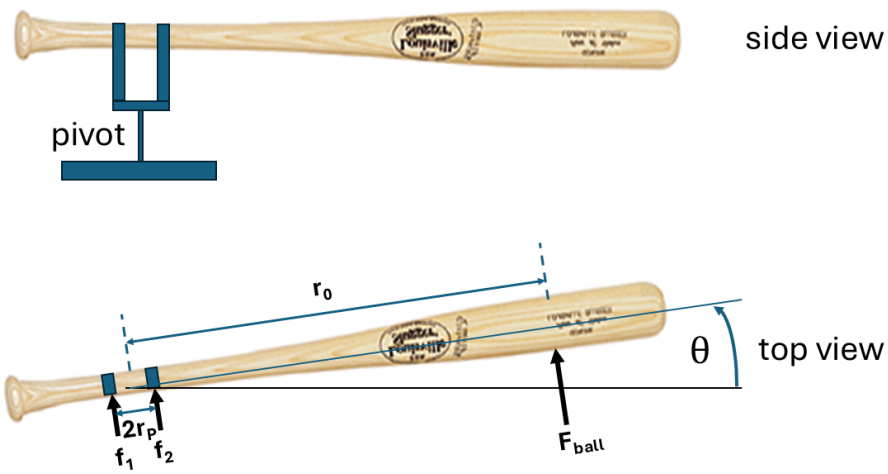

FIG. 2: Side and top views of a bat clamped at two points to a structure that is free to pivot about a vertical axis. The ball exerts force f_{ball} and the clamps exert forces f_1 and f_2 on the bat. All three forces are positive in the directions of the arrows. The top view shows the bat rotated by angle θ about the pivot axis.

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[3] en.wikipedia.org/wiki/Timoshenko

[4] Note that the original matrix **H** is not symmetric.

[5] en.wikipedia.org/wiki/Cholesky Decomposition

[6] William H. Press, *et al.*, *Numerical Recipes: The Art of Scientific Computing* (Cambridge University Press, Cambridge, 1986), Chapter 11 (Eigensystems, specifically subroutines **TRED2** and **TQLI**) and Section 15.1 (Runge-Kutta Method, specifically subroutines **RKDUMB** and **RK4)**.

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